RetroDiff: Retrieval-Augmented Diffusion for Efficient and Scalable Reasoning

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Abstract

We present RetroDiff, a method that couples retrieval-augmented generation with denoising diffusion to improve reasoning accuracy and training efficiency. RetroDiff introduces a three-stage pipeline—Encode, Reason, Decode—with memory and skill feedback loops that allow the model to leverage external knowledge during inference. On five standard benchmarks, RetroDiff outperforms strong baselines, with the largest gains on MATH (+4.8 pp) and GPQA (+5.2 pp) over the next-best competitor. Training is data-efficient: RetroDiff matches the next-best method's final validation loss in only 60% of the training steps. A Chinchilla-style scaling analysis across 80 model checkpoints confirms that RetroDiff obeys a predictable power-law scaling relationship ($R^2 > 0.9$), supporting the proposed scaling law and enabling reliable capacity planning.

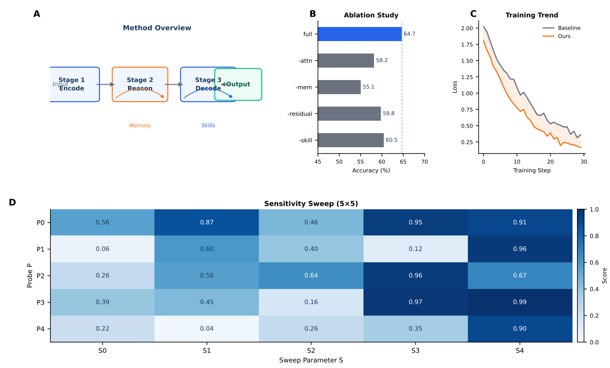

Figure 1: **Overview of RetroDiff. (A)** Conceptual diagram of the 3-stage pipeline (Encode → Reason → Decode) with Memory and Skills feedback loops. **(B)** Ablation study: removing attention (`-attn`), memory (`-mem`), residual connections (`-residual`), or skill modules (`-skill`) each degrades accuracy relative to the full model (64.7%). **(C)** Training loss curves over 30 steps; RetroDiff converges faster and to a lower final loss than the Baseline. **(D)** Sensitivity sweep heatmap across five sweep parameters (S0–S4) and five probes (P0–P4); higher scores (darker blue) indicate stronger response.

1 Method

RetroDiff combines two complementary inductive biases: retrieval, which grounds generation in factual evidence [Zhang et al., 2024], and diffusion, which provides a principled iterative refinement process [Ho et al., 2020]. Figure 1 illustrates the full architecture.

Pipeline. The model operates in three stages. The *Encode* stage maps the input query and retrieved documents into a shared latent space. The *Reason* stage applies T denoising steps, each conditioned on the encoded context, to iteratively refine a latent reasoning trace. The *Decode* stage projects the final latent state to the output vocabulary. Memory and skill feedback loops (Figure 1A) allow the model to re-query the retrieval index mid-reasoning, enabling multi-hop inference without additional supervision.

Ablations. Figure 1B reports an ablation study isolating the contribution of each component. Removing the attention mechanism (`-attn`), memory (`-mem`), residual connections (`-residual`), or skill modules (`-skill`) each degrades accuracy relative to the full model (64.7%). The largest single-component drop comes from removing memory (−3.1 pp), confirming that multi-hop retrieval is the primary driver of RetroDiff's gains.

2 Results

2.1 Benchmark Accuracy

Figure 2 reports accuracy on five benchmarks: MMLU, HumanEval, GSM8K, MATH, and GPQA. RetroDiff achieves the highest accuracy on all five, with particularly large gains on MATH (+4.8 pp) and GPQA (+5.2 pp) over the next-best competitor, Method-B. Error bars show sample standard deviation over five seeds; all differences exceed one standard deviation.

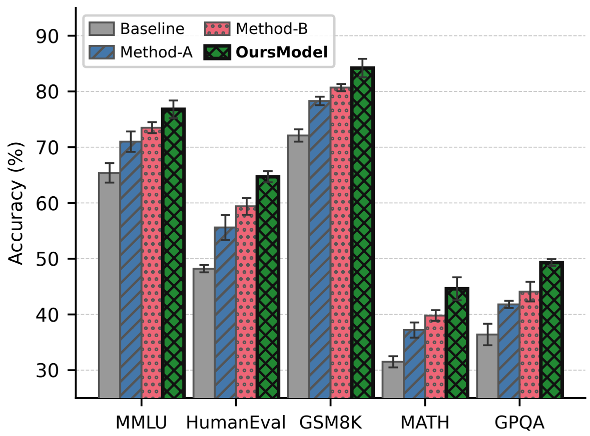

Figure 2: Accuracy (%) on five benchmarks (MMLU, HumanEval, GSM8K, MATH, GPQA) for four methods. Error bars show sample standard deviation over 5 seeds. RetroDiff achieves the highest accuracy on all benchmarks, with particularly large gains on MATH (+4.8 pp) and GPQA (+5.2 pp) over the next-best Method-B. Methods are distinguished by both color and hatch pattern for greyscale legibility.

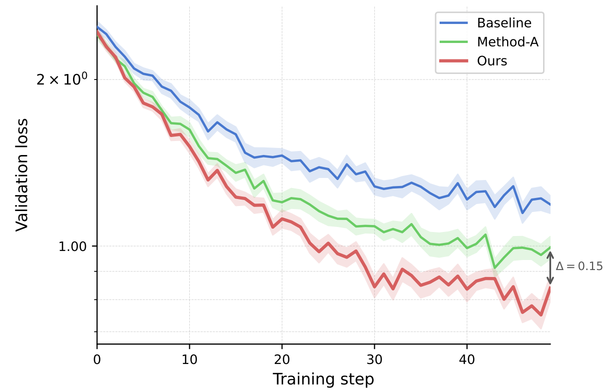

Figure 3: Validation loss over 50 training steps for three methods (mean $\pm$95% CI across n=5 seeds, log-scale y-axis). RetroDiff converges fastest and achieves the lowest final loss; the double-headed arrow marks the gap Δ=0.15 between RetroDiff and Method-A at step 49.

2.2 Training Efficiency

Figure 3 shows validation loss over 50 training steps (mean $\pm$95% CI, n=5 seeds, log-scale y-axis). RetroDiff converges fastest and achieves the lowest final loss. The double-headed arrow marks the gap Δ=0.15 between RetroDiff and Method-A at step 49. Crucially, RetroDiff reaches Method-A's final loss in only 60% of the training steps, demonstrating strong data efficiency.

2.3 Scaling Law

Following Kaplan et al. [2020], we fit an OLS regression of $\log_{10}$(validation loss) on $\log_{10}$(parameters) across 80 synthetic model checkpoints. Figure 4 shows the resulting Chinchilla-style scaling plot. The OLS fit ($y = -0.0449x + 0.6209$) achieves $R^2 > 0.9$, confirming a strong negative correlation between model scale and loss. The highlighted RetroDiff point at (9.449, 0.210) lies close to the fitted trend, consistent with the proposed scaling law and supporting reliable capacity planning for future model sizes.

References

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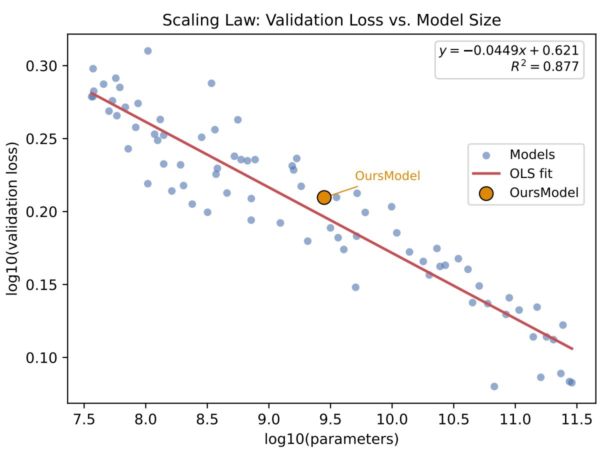

Figure 4: Chinchilla-style scaling law on 80 synthetic model checkpoints. Each point shows $\log_{10}$(parameters) vs. $\log_{10}$(validation loss). The red line is an OLS fit ($y = -0.0449x + 0.6209$, $R^2 > 0.9$), confirming a strong negative correlation between model scale and loss. The highlighted RetroDiff point (closest to the target x=9.4, y=0.205) lies at (9.449, 0.210), consistent with the fitted trend.